

**FEASIBILITY STUDY OF GREAT BUBBLE BARRIER (GBB)
IMPLEMENTATION IN U.S. WATERWAYS**

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1. Introduction

1.1 Background of Plastic Pollution

Plastic contamination of water bodies has become one of the pressing environmental challenges in the world, and this has been motivated by the fact that waste is not effectively handled, and it finds its way to river systems. It is recent estimates that a significant part of 1.15-2.41 million tonnes of plastic are transported annually via rivers into the oceans and therefore rivers present a key source of ocean contamination. In 2020 alone, 1.4 million tonnes of plastic found its way to oceans via rivers, and it is estimated that this number can increase by a significant amount by 2060 unless measures are implemented to control the situation. Considering that rivers are direct forms of transportation, interception in the early stage of river systems will be far better compared to the oceans because during their dispersion and fragmentation into microplastics, besides, ecological damage in the long run will be prevented.

1.2 Problem Statement

Its traditional river cleanup measures and methods remain largely reactive focusing on cleaning up wastes at the lower part rather than reducing transportation of plastic in the upper part. It is estimated that 1.15-2.41 million tonnes of plastic find their way into the oceans through rivers, every year. and it is expected to increase to 3.6 million tonnes by 2060 unless some intervention strategies are adopted effectively. The current methods, like manual collection and mechanical barriers, are restricted by expensive operations, low efficiency, and failure to operate continuously under different hydraulic conditions (Burroughs and Kirk., 2025).

1.3 Aim and Objectives

This paper aims to determine the feasibility of the implementation of the Great Bubble Barrier (GBB) technology in the selected rivers in the United States by evaluating the hydraulic and operational feasibility within the area.

Objectives

- To specifically study measures of important parameters of rivers like river width, which in large rivers of the U.S. may be more than 400800 m, much larger than the 50m operation threshold of GBB.

- To investigate further the flow discharge patterns, where the values tend to be higher than 50000 cubic feet per second at peak conditions, which influence the stability and performance of the system.
- To encompass the comparison of natural rivers to engineered canal systems to determine the best places of deployment.

1.4 Structure of the Report

The report has been structured into nine chapters in order to have a systematised feasibility assessment. The introduction presents the issue, which is backed up by the recent statistics that state that more than 360 million tonnes of plastic waste are produced per year, and about two-thirds of the waste dropped in rivers consists of plastic. The literature review will be based on the existing technologies, and then the methodology and the technical framework.

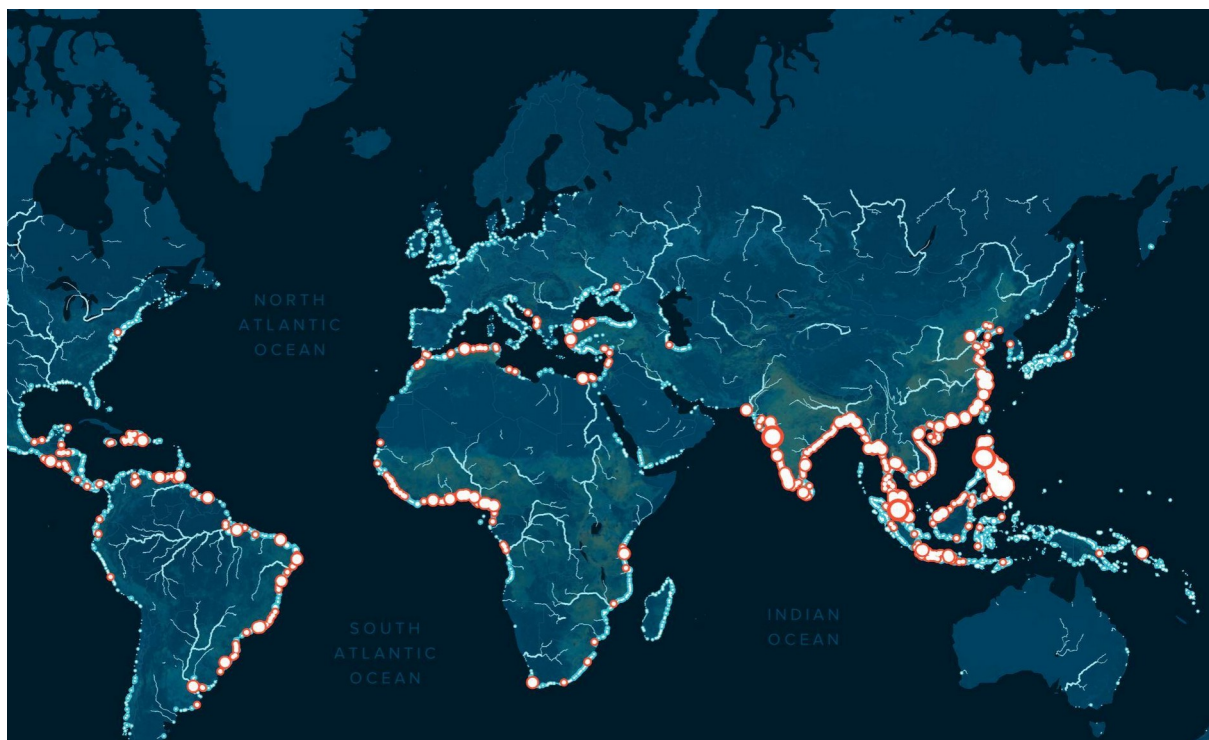


Figure 1 River Plastic Emissions to the world's oceans

(Source: River Plastic Pollution Sources | The Ocean Cleanup (2023))

2. Literature Review

2.1 River Plastic Transport Mechanisms

River systems serve as primary conduits for transporting introducing plastic debris in land to the oceans. It is assumed that rivers carry between 1.15 and 2.41 million tonnes of plastic to the oceans annually and this contributes to the significance of the fluvial processes in the distribution of plastics in the world. Hydrological processes, e.g. surface runoff, flow velocity, discharge rates and seasonal variability, play a major role in this transport.

The entry of plastic waste into the river systems is usually due to improper management of waste, sewage, and the disposal sites being close to the waterways. In the world, about 82 million tonnes of plastic waste are mismanaged annually, and about 5.8 million tonnes of it go into the river systems directly. After the introduction, hydrological processes decide how these materials move and their destiny.

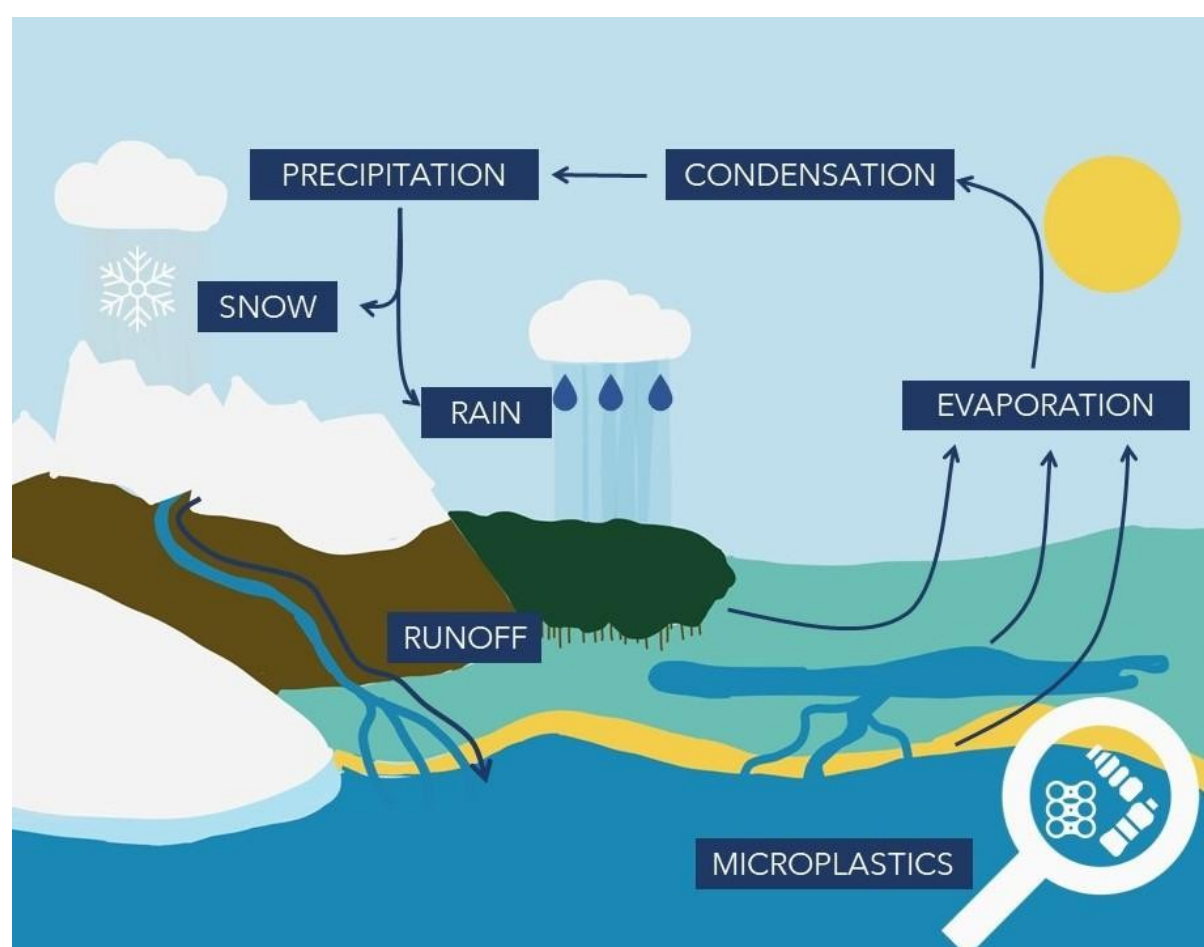


Figure 2 What is plastic rain? Amount of microplastics in the sky.

(Source: Greenwood, K. (2021))

Transport mechanisms are also affected by river morphology and catchment characteristics. Gradients that are steeper and urbanized catchments are conducive to the increased movement downstream, but the vegetation and channel irregularities may trap the plastics. Research has shown that the contribution of over 1,000 rivers to almost 80 per cent of the global riverine plastic emissions shows how the sources of pollution are distributed all over, but not evenly. Also, the bigger the river, the larger the plastic load because of the large discharge and population density in the basin.

2.2 Existing River Clean-Up Technologies

The current river clean-up technologies mostly depend on mechanical interception methods, including floating barriers and skimmers, to intercept plastic waste before it flows to the downstream environments. Mechanical barriers are quite common and usually involve floating booms or angled barriers that are installed across river channels to direct debris on paths to collection points. The systems apply the force of natural flow to focus on floating waste. Recent research has shown that under controlled conditions (e.g. constant flow velocity and concentrated surface-based debris) barrier-based systems with advanced technologies can reach removal efficiencies of up to 85-97% of floating plastic.

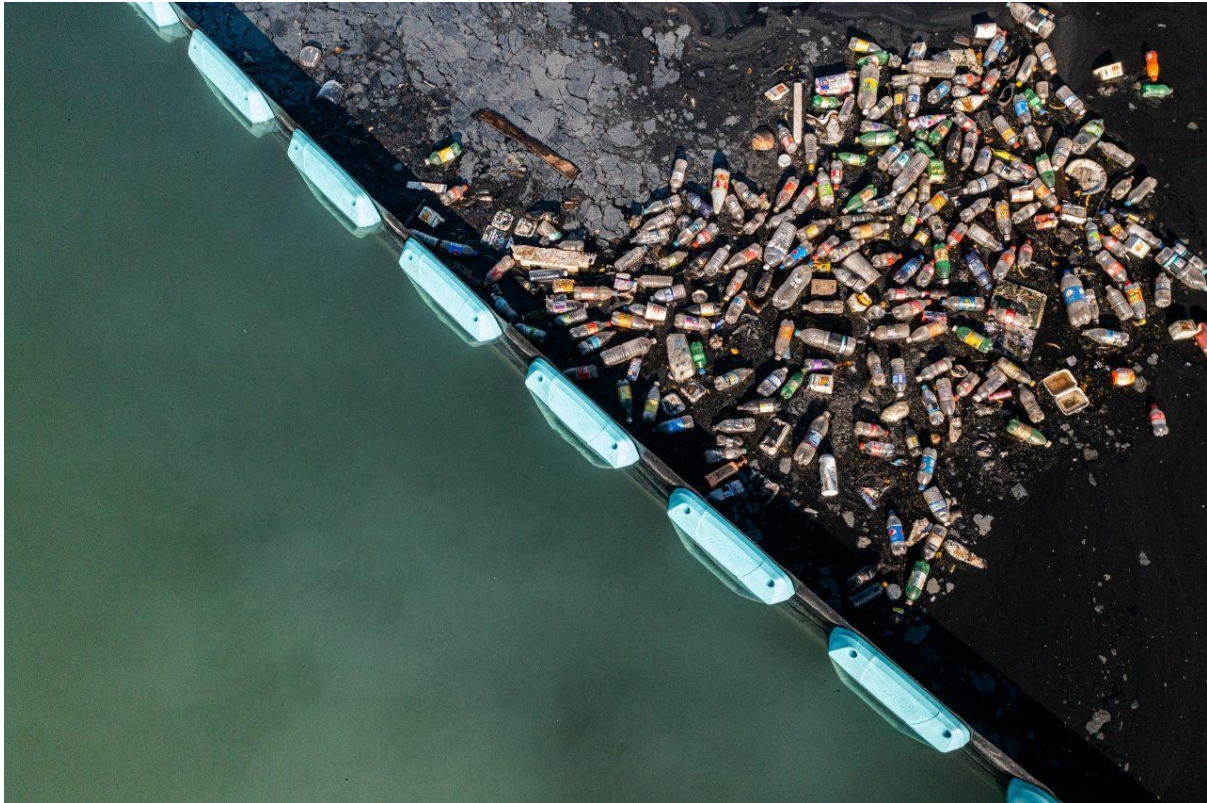


Figure 3 Mechanical river clean-up technologies, including floating barriers and skimmers used for plastic waste interception in urban waterways.

(Source: Correia, R. (2026) Rivers)

The other technology that is highly embraced is skimmers, which are equipment that is designed to actively remove floating debris through suction or conveyor-based harvesters. The modern river skimmer is continuous and can handle large volumes of water. An example of this is that the enhanced skimmer systems have been shown to be capable of processing up to 500,000 litres of water in a year whilst retrieving approximately 20 kg of plastic waste per week. They are efficient, especially in urban waterways where there is localization of waste.

Although both technologies are effective, they have serious limitations. River width, high flow velocities and seasonal changes are examples of mechanical barriers that limit capture efficiency or structural failure (Bogartz., 2025).

2.3 Great Bubble Barrier (GBB) Technology

One more recent form of environmental technology is the Great Bubble Barrier (GBB), which is supposed to trap plastic waste in rivers and deter the entry of plastic waste into the ocean.

The system operates by attaching a perforated tube, at a diagonal with the line of the river bed, to an air compressor. The compressed air is expelled via the tube, creating a continuous bubble column that ascends through the column of water. This upward movement produces a vertical current, which pushes plastic debris up to the surface level and, at the same time, directs it laterally up to a specified collection system.

The diagonal placement of the barrier also plays a crucial role in its functionality since it will use the natural flow of the river to redirect the waste to one side without disrupting the flow of the river or aquatic organisms. It is able to trap plastic as small as 1 mm, and it is able to run continuously without human intervention. It is also wider and deeper in the waterway, hence more comprehensive than the conventional surface-based collection methods.

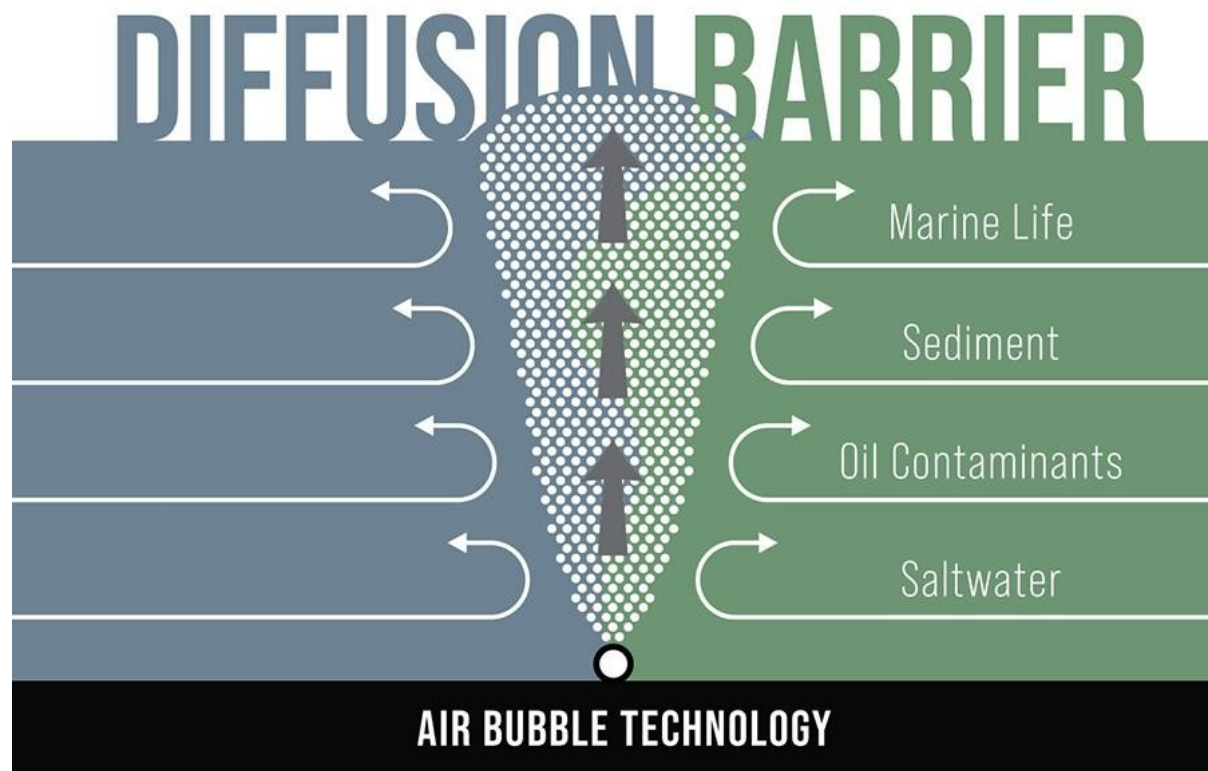


Figure 4 Working principle of the Great Bubble Barrier showing bubble curtain formation and plastic waste diversion towards the collection system.

(Source: Sullair (2023))

The pilot studies have shown that the GBB system is effective based on empirical evidence. An experimental prototype in the River IJssel had an average capture efficiency of about 86% of plastic debris in controlled conditions. Also, the original long-term installation in

Amsterdam, which has been in operation since 2019, demonstrated stable outcomes, as approximately 190,000 pieces of plastic are captured every year (around one tonne of waste). The monthly observation also shows that the average monthly collection is approximately 80 kg of garbage, with plastics making up more than 70% of the garbage.

2.4 Research Gap

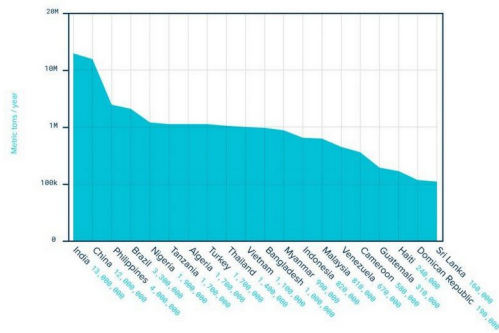
Although the global concern about the problem of riverine pollution with plastic is gaining momentum, the lack of quality, feasibility-based research is high in the case of the United States. The literature at hand quantifies plastic transport globally, within the range of 1.15-2.41 million tonnes of plastic entering oceans per year through rivers. With 70-80 per cent of the total coming on land. These researches however, are mostly concentrated in the Asian and developing regions where most of the high-emitting rivers are found. According to recent discoveries, about 1000 rivers cause almost 80 per cent of the total plastic emissions worldwide, but U.S. rivers are not fairly represented in these studies.

FROM LAND-BASED MISMANAGED PLASTIC WASTE TO RIVERINE PLASTIC EMISSIONS



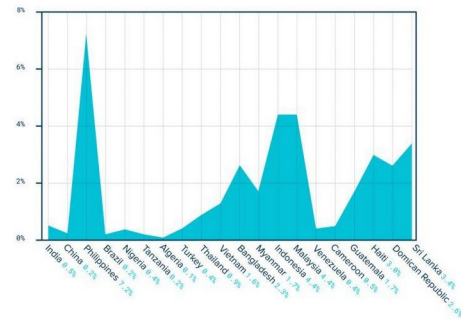
1 MISMANAGED PLASTIC WASTE

METRIC TONS / YEAR



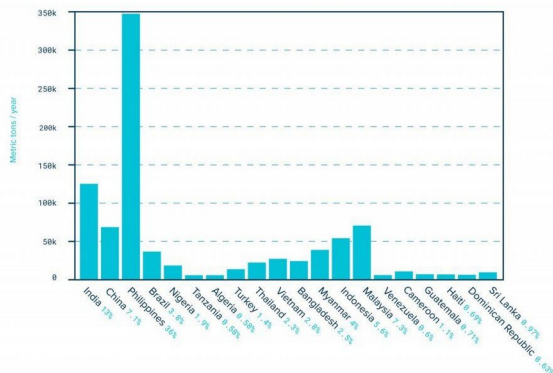
2 AVERAGE EMISSIONS PROBABILITY

P(E) [%]



3 PLASTIC EMISSIONS M_E

PER COUNTRY IN 2015



THE TOP 1000 RIVERS EMIT
20 MILLION
GARBAGE BINS
OF 240L PER YEAR

THAT'S ON
AVERAGE
20,000 BINS
PER RIVER



- MISMANAGED PLASTIC WASTE**
Land-based mismanaged plastic waste generation is calculated by quantifying the waste management practices, consumption patterns, and population density of a particular country or region.
- AVERAGE EMISSION PROBABILITY**
The probability for mismanaged plastic waste to reach a river and subsequently the ocean depends on rain, wind, land use, terrain slope, and the distance to the nearest river and ocean.
- RIVERINE PLASTIC EMISSIONS**
To calculate the riverine plastic emissions within a country, mismanaged plastic waste is multiplied by the emission probability on a ~90 x 90 meter spatial resolution.

Figure 5 Global distribution of riverine plastic emissions highlighting major pathways of plastic transport from land to oceans.

(Source: erika (2021))

Further, recent analysis dwells upon the measurement of the pollution, rather than on the technological feasibility, particularly in the example of the Great Bubble Barrier (GBB) (Politano *et al.*, 2024, August). Little research has been done on the impact of hydraulic variability, channel geometry and flow stability of the U.S. rivers on system performance. This loophole underscores the necessity of location-specific feasibility studies that synthesize engineering standards with environmental standards.

3. Methodology

3.1 Research Design

The paper uses a quantitative feasibility analysis as the methodology to examine the suitability of the chosen waterways in the United States to implement the Great Bubble Barrier (GBB). The methodology involves the methodical evaluation of hydraulic parameters, such as the width, depth, velocity of the flow, and stability of the water level in the river. These variables are important since the dynamics of river flow have a direct impact on the transport and capture of plastics. Recent research has shown that rivers constitute major routes of plastic pollution, and the majority of yearly plastic transportation is carried out in times of high flow, like floods, which reveals the significance of velocity and discharge variability in the feasibility study.



Figure 6 Measurement of river flow velocity and hydraulic parameters used in quantitative feasibility analysis.

(Source: Nivus (2026))

3.2 Selection of Study Areas

Quantitative hydraulic criteria were used to select the areas of the study so as to determine the viability of the implementation of the Great Bubble Barrier (GBB). The rivers were chosen to

account for the different physical and flow conditions found in the U.S. waterways. Some of the important parameters were channel width, channel depth, flow velocity and variability in season. As an example, the Susquehanna, being a large river, is often more than 300-400 m wide, which is many times greater than the working limit of 50 m, and thus cannot be installed directly. On the same note, natural river velocities tend to be near or even higher than the velocity of 1 m/s in normal or flood conditions, which is higher than the necessary limit of 0.5 m/s to have stable bubbles (Avery *et al.*, 2025).

River Cross Section Measurements

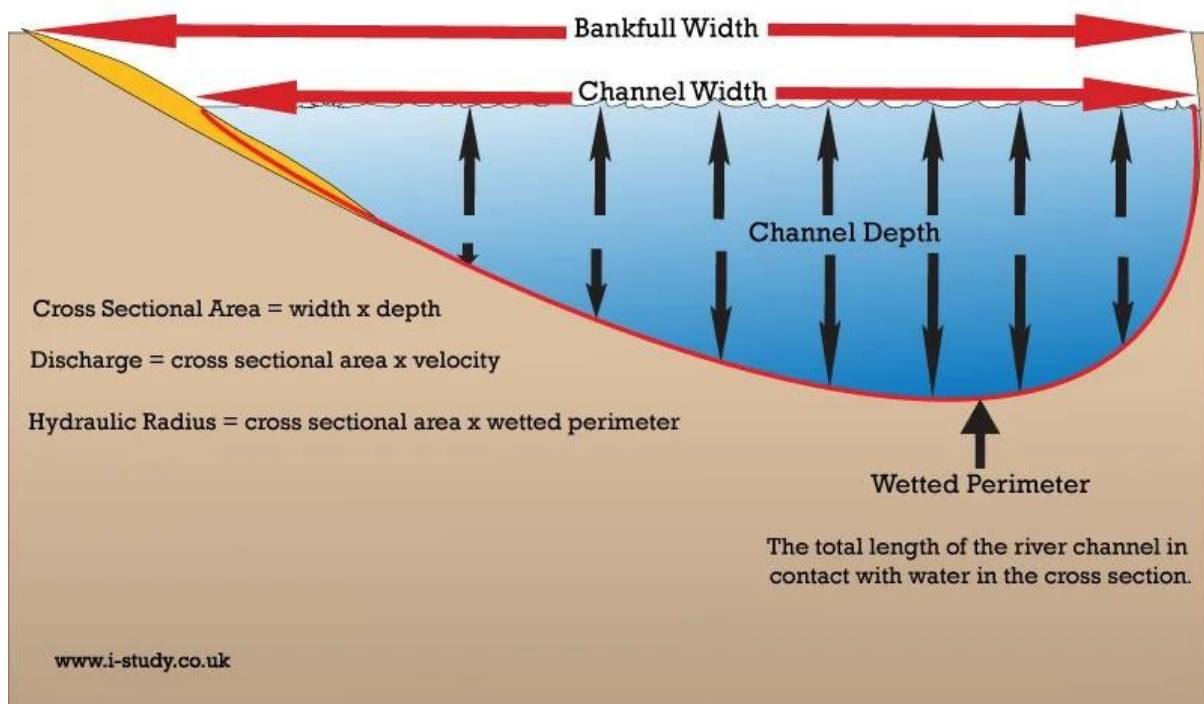


Figure 7 Key hydraulic parameters influencing river flow, including channel width, depth, and velocity, which are critical for assessing the feasibility of GBB implementation.

(Source: Rivers (2022))

3.3 Data Sources

The investigation is based on the secondary hydrological data acquired at the United States Geological Survey and related monitoring systems. USGS stream gauge stations give real-time and past data of the discharge, water level and flow of rivers. An example of this is the recent

measurements of the Schuylkill River at Reading, which indicated a mean streamflow of about 2,370 cubic feet per second (cfs), but the historical mean was about 2,556 cfs, which shows that the flow conditions varied moderately.

Other data sets are hydrological reports with measurements of drainage area, including the Schuylkill River basin with an approximate area of 1,760 square miles, seasonal changes of flow and gauge height. These data sets enable evaluation of the key feasibility parameters like the stability of velocity, uniformity of depth and fluctuation of discharges.



Figure 8 Example of a USGS stream gauge station used for measuring river discharge, water level, and flow velocity for hydrological analysis

(Source: Ask The Man (2021))

3.4 Feasibility Criteria

The possibility of installing the Great Bubble Barrier (GBB) is identified based on the principal hydraulic parameters, such as the width, depth, velocity of flow, and the stability of water

levels of the river. To work efficiently, the river should not be wider than about 50 m to make sure that the bubble curtain is able to cover the channel. This required depth of not less than 1 m, although engineered waterways, such as the Chicago Sanitary and Ship Canal, were quite deep, between 827 ft (2.428 m), and this depth permits formation of stable bubbles.

The flow velocity is a key parameter because the maximum performance of GBB is achieved at less than 0.5 m/s (Press and Siqueira., 2022). Under normal conditions, velocities in controlled canal systems vary between 0.08 and 0.15 m/s, which is a good range of velocities that allow efficient waste guidance, whereas faster velocities in floods decrease effectiveness.

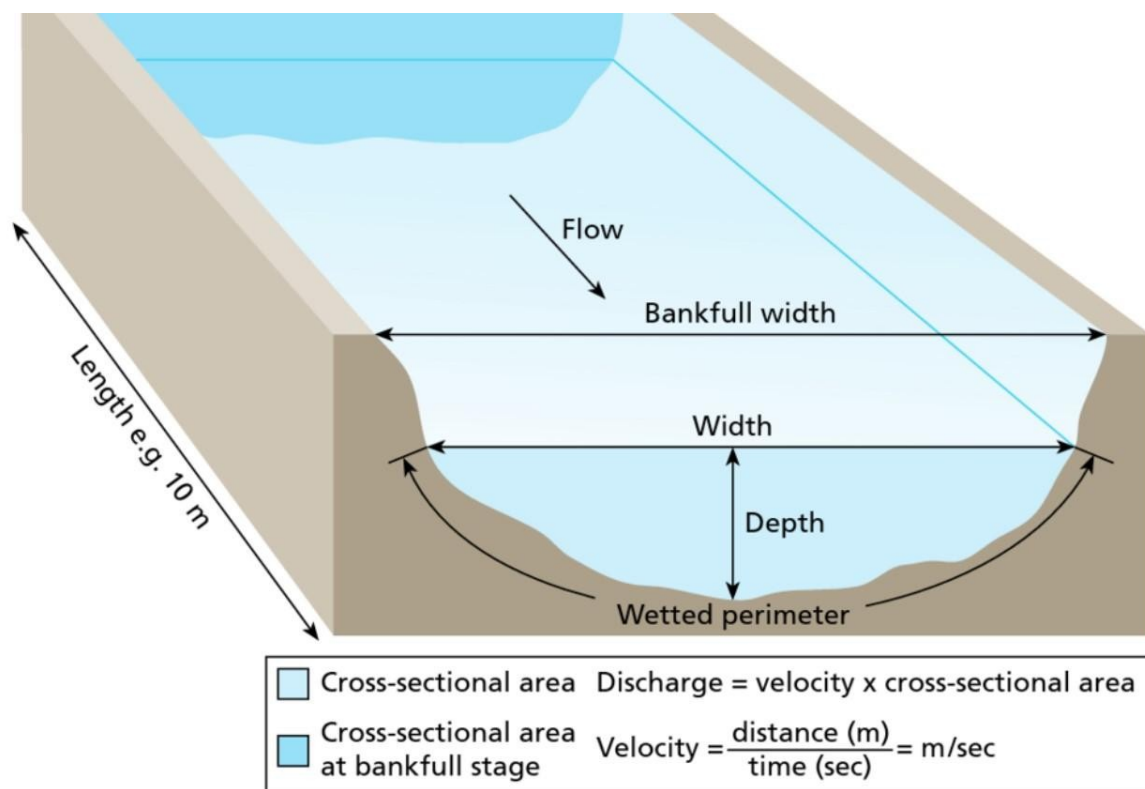


Figure 9 Key hydraulic parameters influencing GBB feasibility, including river width, depth, flow velocity, and channel stability.

(Source: HAPI PROJECT (2018))

Lastly, water level fluctuation should not exceed ± 1 m to ensure that the system is stable. High variability, which is typical of natural rivers, also causes disruption of the alignment of bubbles and a decrease in plastic interception efficiency, which limits feasibility.

4. GBB System Design and Technical Requirements

4.1 System Components

The Great Bubble Barrier (GBB) system has three major components, namely, an air compressor unit, a perforated diffuser tube and the resulting bubble curtain. The air compressor feeds constant pressurised air, usually at industrial levels of about 400 V three-phase power, which maintains constant and uninterrupted air flow during the 24 hours of operation. Diffuser systems typically work between 1 and 10 m³/h per unit, making the volumetric airflow rate a critical parameter and usually depending on the river width and coverage required.

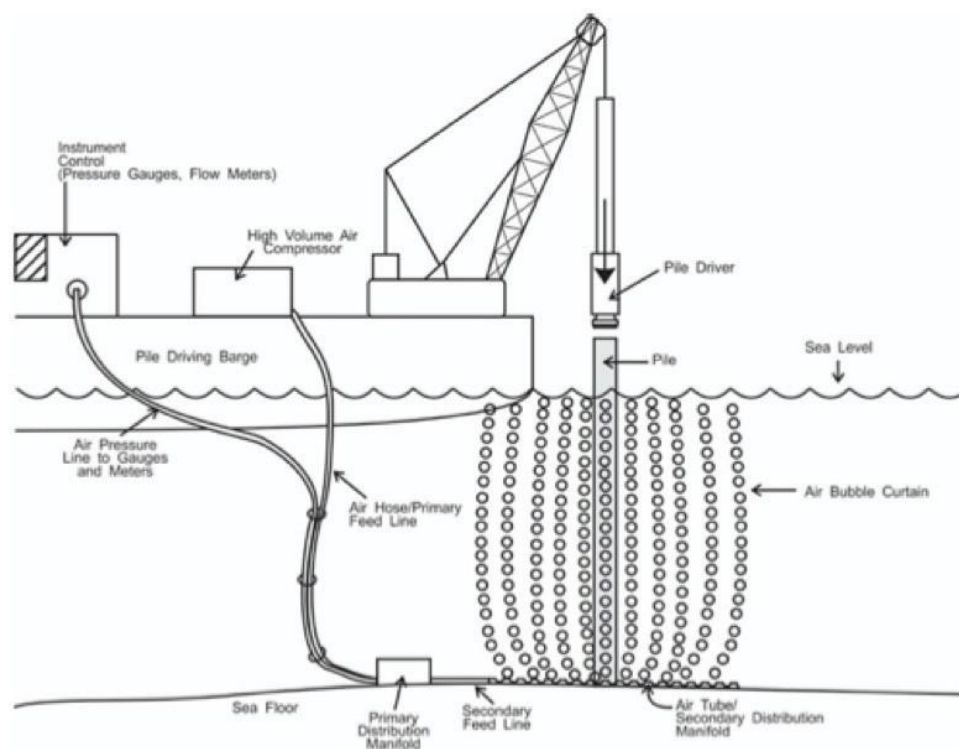


Figure 10 Schematic representation of the Great Bubble Barrier system showing the air compressor, perforated diffuser tube, and formation of the bubble curtain guiding plastic waste toward the collection zone.

(Source: Innovative Fish Deterrent System Will Continue to Minimize Potential... (2020))

Whenever the compressed air flows through the perforations, it forms a bubble curtain that forms small bubbles of a diameter of about 0.8-3 mm each. These bubbles are blown up, resulting in a vertical flow field whereby plastic waste will be carried to the surface by the up velocity. Bubble flux is a control of system efficiency, which is given as:

$$Q = A \times V$$

Q is the rate of airflow, A is the area of diffusers, and v is the air velocity. This combined system will allow the interception of waste continuously with energy efficiency.

4.2 Operational Mechanism

The working principle of the flow is identified under multiphase flow as it is the Great Bubble Barrier (GBB), where the river has compressed air flowing into the riverbed through a perforated pipe. The injected air forms a vertical column of rising bubbles generating a field of upward velocity that transfers plastic particles up to the water surface. (Glickstein., 2022). This natural aspect of the river flow gives the necessary support to this buoyancy-aided motion, and the river flow is usually less than 0.5 m/s to achieve maximum efficiency in the waste transportation in a lateral fashion.

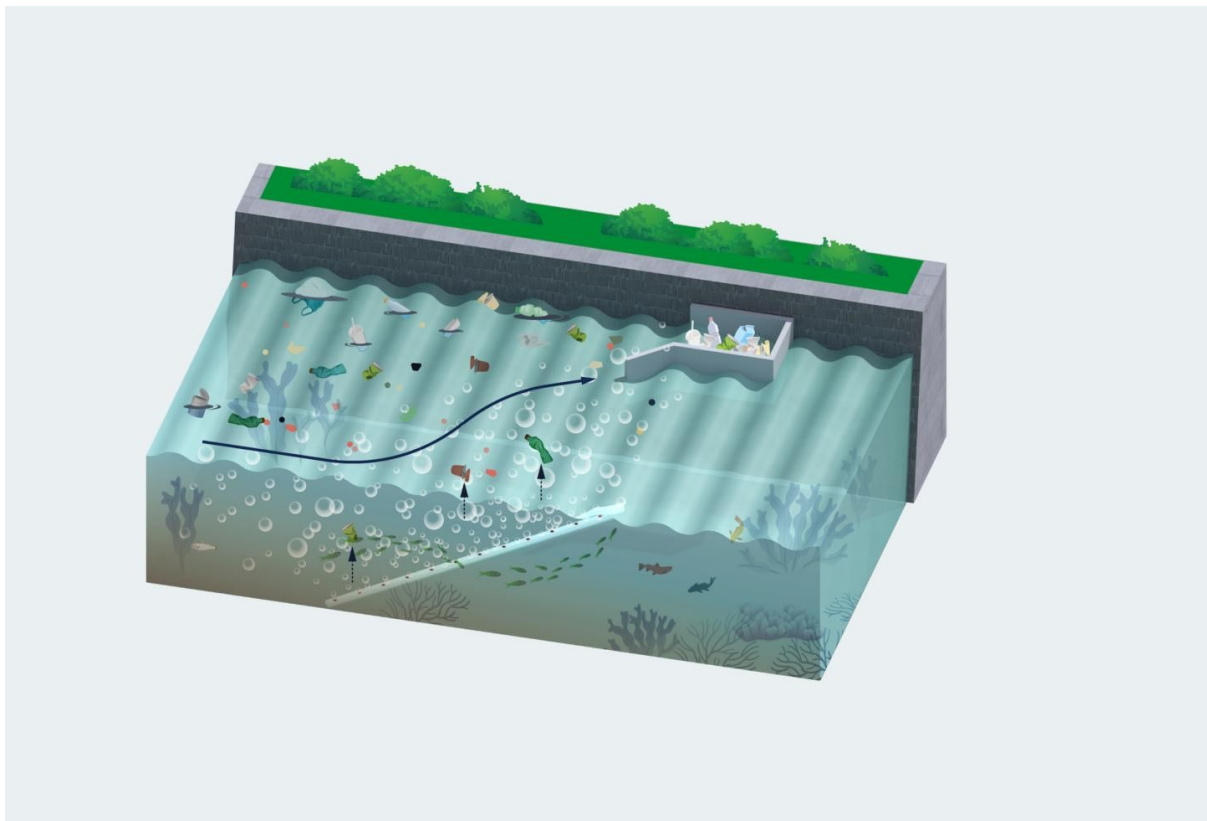


Figure 11 Operational mechanism of the Great Bubble Barrier showing vertical bubble plume, diagonal waste transport, and surface collection system.

(Source: Dialogue Earth (2021))

The horizontal flow of a river and vertical bubble velocity produce the effect of a diagonal flow, which pushes floating debris to a collection mechanism. According to experimental studies, bubble curtain guidance efficiency varies between about 20 per cent at high velocities (0.4 m/s) and 47 per cent at low velocities (0.1 m/s), and shows the performance to be sensitive to the conditions of the flow.

Air flux per unit width can be used to describe the hydrodynamic behaviour:

$$q_{air} = \frac{Q^{air}}{W}$$

Q^{air} is the rate of air flow, and W is the width of channels. The parameter regulates the strength and stability of plumes.

Moreover, the growing bubbles cause recirculation areas and shear flow, which create an effective barrier to short-cut suspended particles without hindering navigation.

4.3 Installation Requirements

The Great Bubble Barrier (GBB) was installed, requiring certain conditions of the site, which are necessary to guarantee the best performance and stability of the structure. The system is commonly installed in waterways with a width of 3050 m and a depth of more than 1 m (in order to ensure that the bubble curtain can fully develop vertically). Canals that are engineered with a depth of 2-8 m are deemed ideal since the hydraulic conditions remain stable. The velocity of the flow should not exceed around 0.5 m/s to ensure efficient rise of the bubbles and lateral deflection of floating debris, since faster flow interferes with the development of the plume.

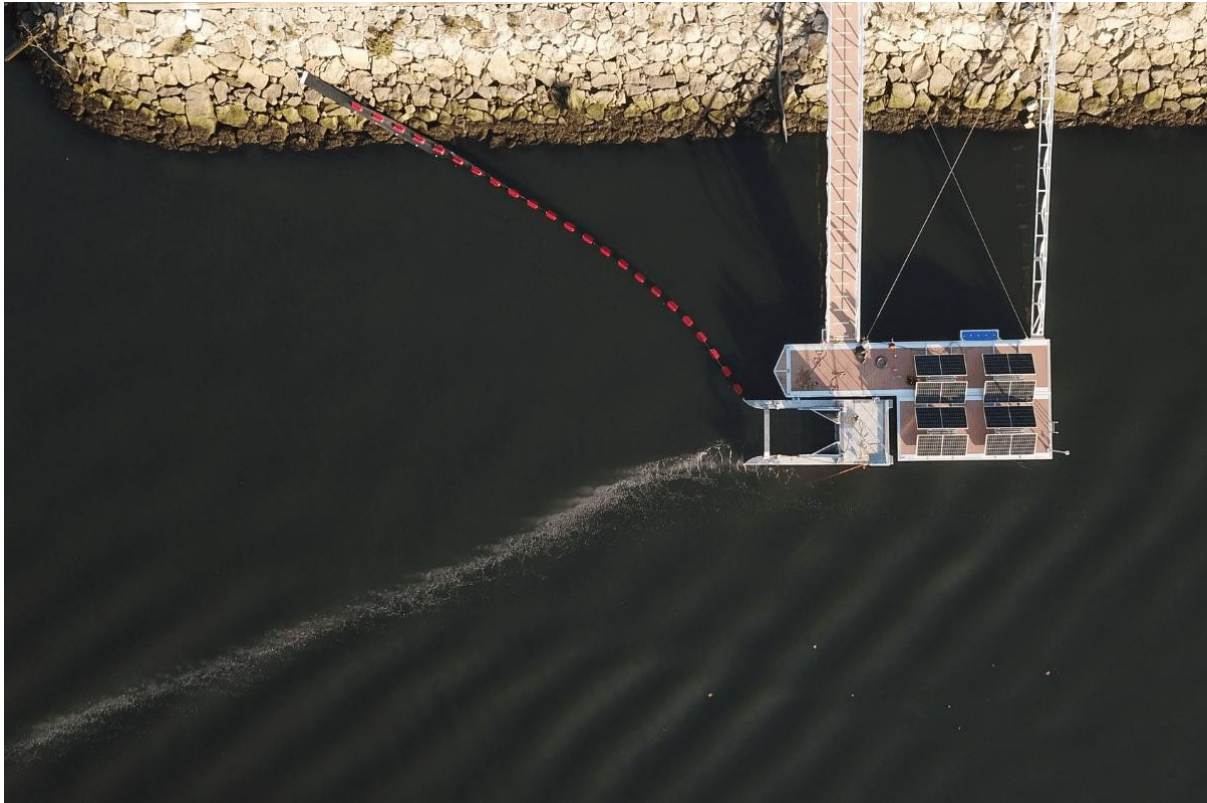


Figure 12 Schematic representation of Great Bubble Barrier installation showing perforated pipe placement, airflow generation, and diagonal bubble curtain guiding plastic waste toward the collection point.

(Source: The Great Bubble Barrier® (2026))

The airflow rate required for bubble generation can be estimated using:

$$Q = A \cdot v$$

and Q is air discharge (m^3/s), A is perforation area, and v is injection velocity.

Other limitations are a stable water level (less than 1 m change), lack of excessive sedimentation, and the safety of perforated pipes to the riverbed.

5. Initial River Analysis

5.1 Susquehanna River

The Susquehanna River is proven to be one of the biggest river systems within the United States, with its basin of about 27,500 square miles and a length of about 444 miles. It contributes a lot of freshwater and pollutants into the Chesapeake Bay, providing almost half

of the entire inflow to the bay. The river poses significant constraints to the installation of the Great Bubble Barrier (GBB) system despite its significance to the environment.

Hydraulically speaking, the river is larger than the critically required width of 50 metres in most places, and some sections are well over 300 metres. Moreover, the flow discharge also differs considerably, and average discharge rates are about 40,670 cubic feet per second and maximum flow surpassing 1,130,000 cubic feet per second in extreme events. This great variability results in instability in flow velocities, mostly exceeding the minimum operating safe bubble velocity, 0.5 m/s. Although side-channel locations, e.g. around Sunbury, might serve dimensional requirements well, the contribution made to the total waste transfer is low, decreasing the efficiency of interception.



Figure 13 Wide channel morphology and variable flow conditions of the Susquehanna River demonstrating hydraulic challenges for GBB implementation.

(Source: Coluccio, K. and Morgan, L.K. (2019))

5.2 Schuylkill River

The Schuylkill River is a river in Pennsylvania that is around 135 miles in length and also covers a watershed of close to 1,900 square miles; therefore, it is an important tributary of the Delaware River system. The river also has a significant variability of the flow conditions as the discharge values have been recorded to range between the bare minimum of about 995 cubic feet per second (cfs), the maximum of over 40,000 cfs and the average of 2,875 cfs. The latest data on monitoring have shown that the flow varies by approximately 3,000 cfs, with short-term averages of almost 4,600 cfs, thus showing continued hydrological variability.

Optically, the river does not satisfy the critical operation requirements of the Great Bubble Barrier (GBB) system. The heavy discharge variation causes fluctuation in flow velocities, especially in conditions of floods when water strength is high, resulting in disruption of bubble formation and control towards direction (Zhou *et al.*, 2025). Moreover, gauge height changes between a level of about 6.5 feet during usual conditions and more than 11 feet during flood periods show that there was a lot of instability in the water level.

Narrower portions, like Manayunk Canal, have something contained in narrower widths (about 1824 m), but these channels are not the main discharges of the river. Subsequently, a high percentage of floating plastic debris would miss the areas impacting greatly on interception effectiveness.

Thus, even though a substantial part of the city's waste transport is served by the river Schuylkill, this river cannot be considered as the location of the GBB development because of the lack of hydrological stability, ineffective waste interception, and uneven operational conditions.

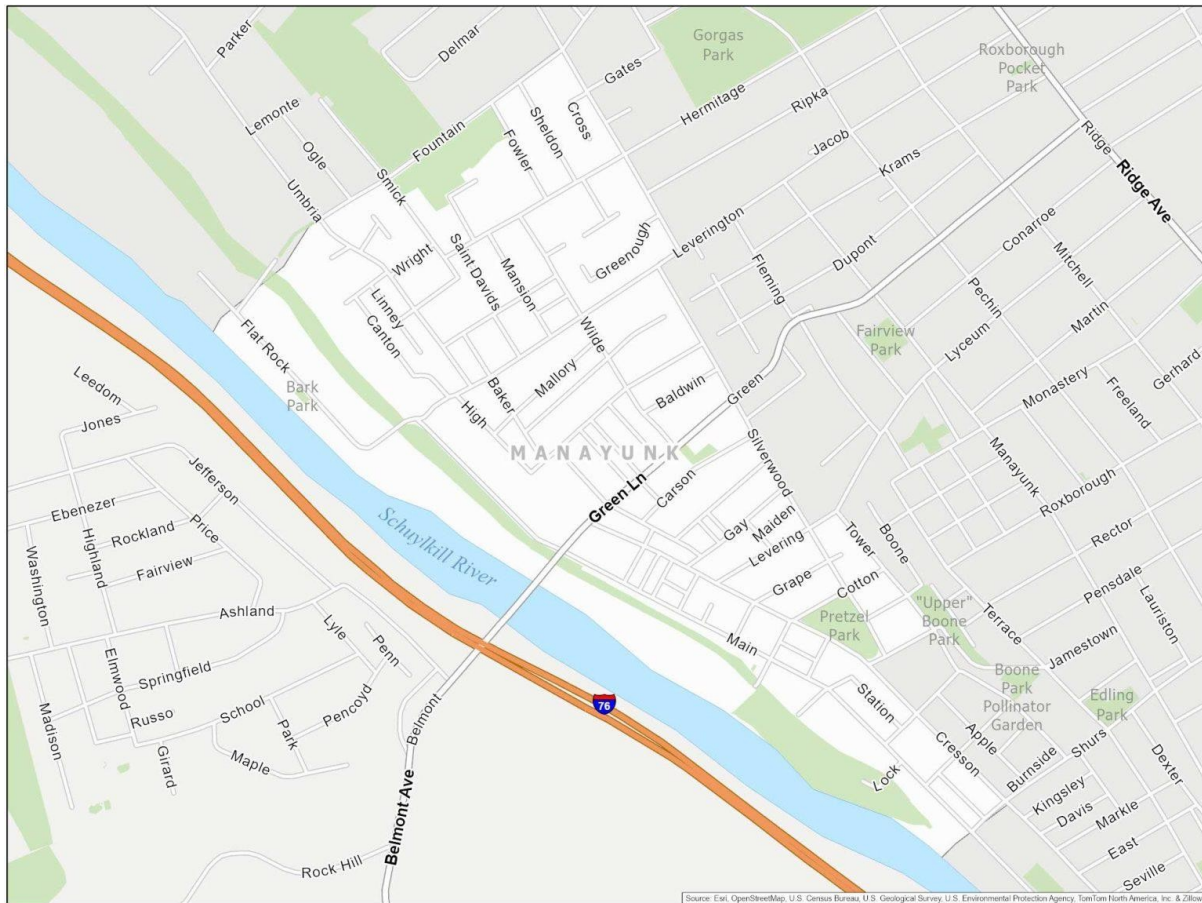


Figure 14 Aerial comparison of the Schuylkill River main channel and the adjacent Manayunk Canal, illustrating the contrast between wide, high-discharge river flow and the narrow, controlled canal system.

(Source: GISGeography (2023))

5.3 Lehigh River

The Lehigh River is highly hydrologically erratic, so it cannot be used in the deployment of the Great Bubble Barrier (GBB). The river is about 175 km long and typified by seasonal variations of discharge and water level. Hydrological records reveal that the flow rates may come in a very broad range and could be as low as less than 50 cubic feet per second (cfs) during the dry summer seasons to a high of more than 20,000 cfs during the storm and flood times. This variability has a direct impact on the velocity of the flow, which is most of the time higher than the GBB operation limit of 0.5 m/s in the case of high discharge rates.

On the channel geometry, the river is 30 to more than 100m in width at various points, which is often even higher than the advised maximum width of 50 m to ensure optimal bubble barrier

performance. Moreover, the depth of water is uneven, where shallow areas drop below the minimum limit of 1 m in times of low flow.

Seasonal aspects also make matters complicated in terms of feasibility. Ice may develop in the winter seasons, whereas increased rainfall activities may enhance the transportation of sediments and the turbulence, which disrupts the ability of such bubbles to remain stable and effectively intercept the wastes.

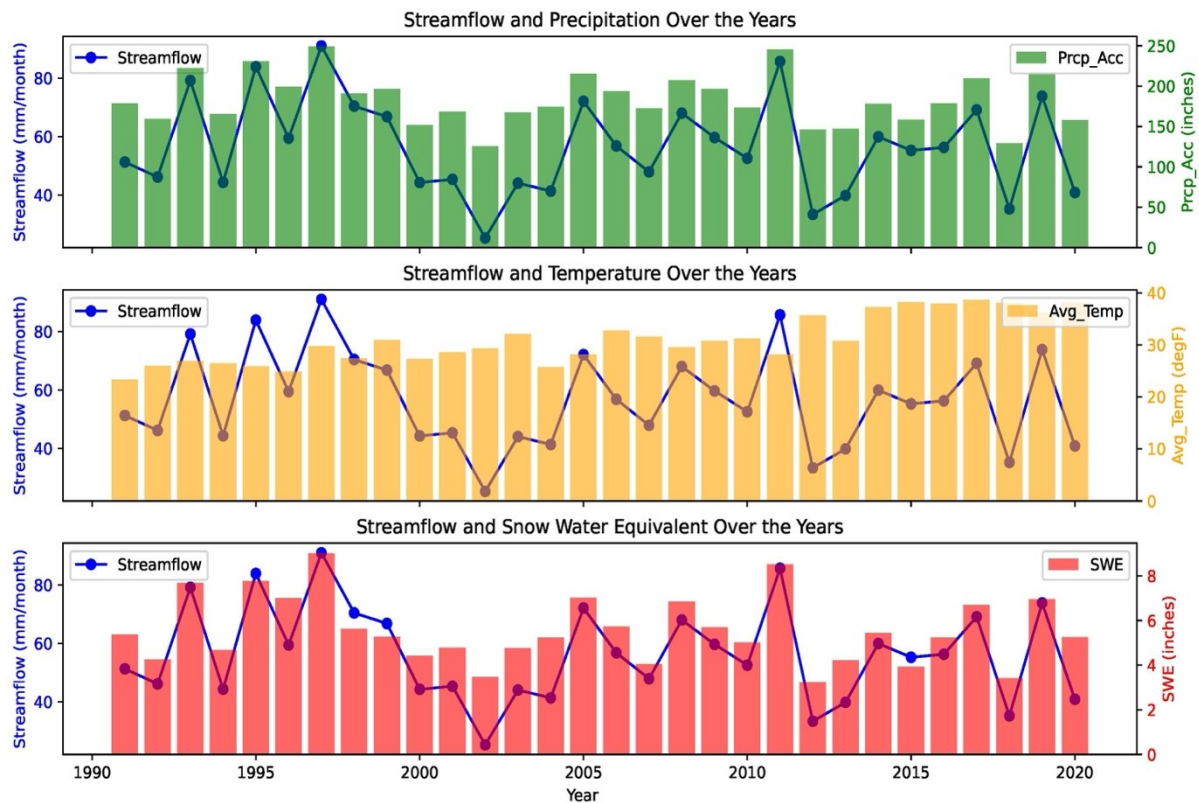


Figure 15 Seasonal variation in flow conditions of the Lehigh River, illustrating significant fluctuations in water level and discharge that affect hydraulic stability and feasibility of GBB implementation.

(Source: Thota, S. et al. (2024))

In general, the Lehigh River does not pose a stable condition to fulfill the hydraulic and operational requirements of efficient implementation of GBB.

6. Multi-State River Feasibility Exploration

6.1 California – Los Angeles River

The case of the Los Angeles River, California, is one of the most altered urban water systems that might seem fit at first glance as the site where the Great Bubble Barrier (GBB) can be implemented because of the channelized system used there. The river stretches up to about 51 miles and is defined by a channel, which is concrete-lined and whose entire purpose is not to fulfil an ecological or waste interception aspect, but flood control.

The geometry and the flow conditions in the river are highly variable hydraulically. Downstream channel widths normally vary between 120 and 180 metres, more than the standard 50 metres to ensure efficient operation of GBB. Also, although the average discharge is comparatively low at only around 226 cubic feet per second, during storm occurrences the peak flows may be as high as 129 000 cubic feet per second; this is a very extreme variation range.



Figure 16 The Los Angeles River showing a concrete-lined channel designed for flood control, illustrating the wide and highly engineered structure that limits the feasibility of bubble barrier implementation.

(Source: Los Angeles River)

Moreover, the river does not have regular depth profiles, with most parts of the river being shallow in dry periods but having extreme shoot-ups of depths during high-flow periods. Turbulence control, measured by the existence of a concrete channel, also shortens the capacity to form bubbles and direct flow by which the plastic interception is attained.

Thus, the Los Angeles River does not fit the major GBB feasibility requirements, especially with regard to width, flow consistency and seasonal changes, which make it inapplicable in terms of reliable and constant use.

6.2 Texas – Trinity River

The Trinity River in Texas is one of the major river systems entirely situated within a single state of the U.S., and it has a total length of about 710715 miles long with a drainage area of about 18000 square miles. Although large-sized with great importance in the transport of freshwater and sediments to the Galveston Bay, the river has a high degree of hydrological variability, thus making its implementation on the Great Bubble Barrier (GBB) less suitable.

The river exhibits extremely discursive flow patterns and an annual discharge average of millions of acre-feet, but is very patchy as a result of the concentration of rainfall and the releases of reservoirs (Elliott *et al.*, 2024, July). As observed in the field, there is a wide range between the discharge levels of a couple of hundred cubic feet per second at low-flow periods up to more than 20,000 cubic feet per second at peak flow conditions, which means that the variability is extreme. This fluctuation causes inaccurate flow velocities, which at times are much higher than the ideal velocity of 0.5 m/s to achieve successful bubble formation.



Figure 17 Aerial view of the Trinity River in Texas, illustrating its wide channel, floodplain expansion, and variable flow conditions that limit the feasibility of controlled barrier-based waste interception systems.

(Source: A look at Trinity River in Dallas during Flood Warning (2018))

Furthermore, the Trinity River is subject to massive flooding, the history of which has made it many times wider, reaching a width of 1.5 miles in extreme situations. Wide channel geometry and transport of sediment further decrease system efficiency even in normal conditions.

Consequently, in spite of the sufficient depth conditions averaging at 30 feet, the joint impact of the flow instabilities, over width, and the risks of floods makes the Trinity River unreliable and ineffective to implement GBB on a regular and effective basis.

6.3 New York – Hudson River

The Hudson River is a significant stream within the state of New York and is about 315 miles in length, and a tidal estuary near the lower part. Its hydraulic properties introduce high challenges to the deployment of the Great Bubble Barrier (GBB) system (Lee *et al.*, 2024). The river is also of a significant variability of width, where it varied to a maximum of 5 kilometres (3 miles) in certain sections, including Haverstraw Bay, and to some extent only 300 metres in the upper part. Moreover, the Hudson River has severe tidal activity, and every six hours or so, the water flow is reversed with the oceanic tidal processes.



Figure 18 Aerial view of the Hudson River illustrating its wide estuarine structure and tidal influence, which results in bidirectional flow and limits the feasibility of stable GBB implementation.

(Source: Wikipedia Contributors (2019) Hudson River)

Depth conditions, as a rule, are adequate, between 10 and 15 metres on average and over 60 metres in extreme points, and do not mitigate the lack of stability of flow behaviour. Moreover, there is a major variation in the level of discharge whereby the amount of fresh water discharging at the river mouth per second is approximately 21,900 cubic feet, which implies great hydraulic power that can interfere with plastic nature guidance and the formation of bubbles.

As a result, the Hudson River, although vital as a significant transport channel in waste management, does not satisfy critical GBBs operation requirements, especially with regard to stability in flow and direction, which makes it impractical and unapplicable in real-time cleaning.

6.4 Multi-State – Mississippi River

The current river system, the Mississippi River, stretching over ten U.S. states, is one of the biggest and most complicated river systems in the world, covering about 41% of the total

territory of the United States. Although it is important in the transportation of a huge amount of water and waste, its hydraulic nature renders it unsuitable to apply it in the Great Bubble Barrier (GBB). The river has very high discharge levels, where the mean discharge is about 593,000 cubic feet per second and maximum amounts are over 3,000,000 cubic feet per second, which is significant variability of flow and intensity.



Figure 19 Aerial view of the Mississippi River illustrating its extensive width and large-scale flow, highlighting the hydraulic challenges associated with implementing bubble barrier systems.

(Source: How the Water Shapes Us)

The river, also, is often much wider than a few hundred metres and in parts goes close to one mile, far above the requirement of the GBB of an operationally maximum width of 50 m. The river is also very deep, with the depth difference being usually 9-15 feet in the upper region to more than 200 feet at its lower boundaries, which is highly spatially inconsistent. More so, the changing seasonal variation of discharge and recurring floods lead to varying flow patterns, which to a significant point is less feasible to sustain a consistent bubble curtain system. Under these conditions, the Mississippi River does not satisfy major GBB requirements, especially concerning the width, stability of flow, and control of velocity.

6.5 State-wise Agency Analysis

In the United States, environmental governance is based on a multi-level institutional organizational functioning of the federal and state agencies. At the federal level, the most significant actor in controlling the water quality, pollution, and compliance with the environment is the United States Environmental Protection Agency. By 2025, the budget size of the EPA will be about 9.1 billion dollars each year, and it will be staffed by over 14,000 employees, which will help conduct nationwide environmental supervision and impose laws.

As an example, the California Environmental Protection Agency, Texas Commission on Environmental Quality and the New York State Department of Environmental Conservation have the task of implementing the federal regulations and overseeing the local water bodies. These agencies control permitting programs, monitoring of pollution and restoring rivers in their jurisdictions.

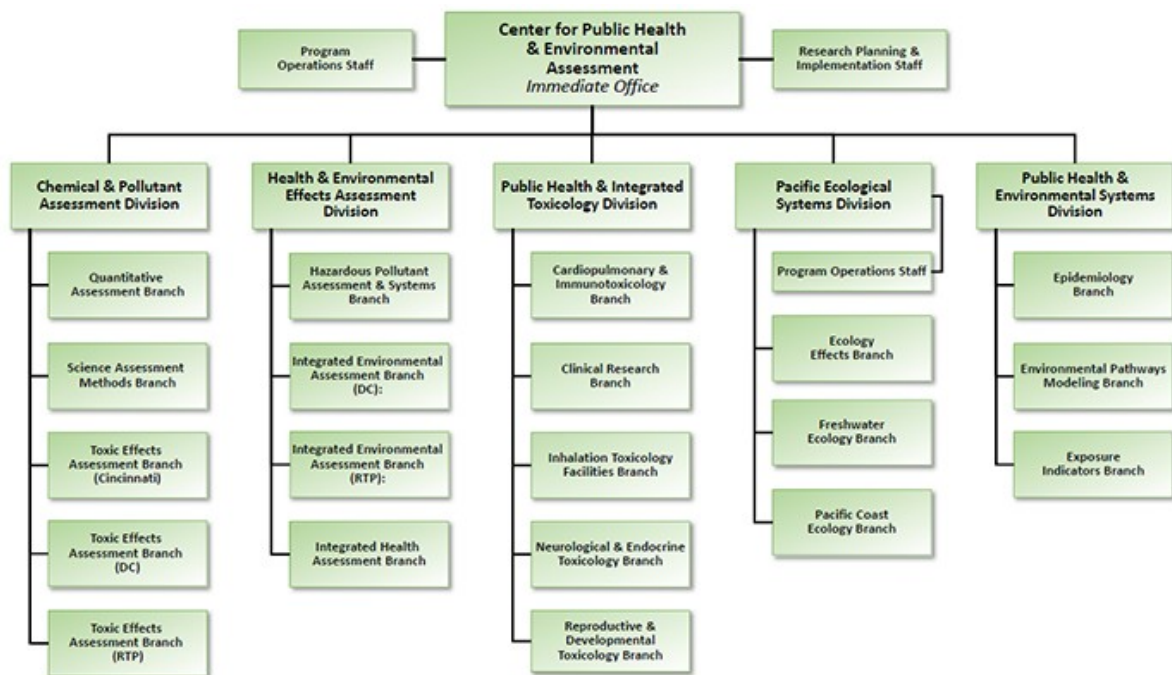


Figure 20 Multi-level environmental governance structure in the United States, illustrating the coordination between federal (EPA) and state-level environmental agencies in managing water resources and pollution control.

(Source: US EPA,OA (2019))

This type of collaborative governance form enables regions to be managed in a region-specific manner, but the regulation at the national level remains constant. But differences in revenue and technical capability and enforcement methodology between states may mean that the large-

scale environmental intervention like the Great Bubble Barrier (GBB) does not work well in challenging and highly dynamic river systems, especially in the Great Divergent Rivers.

7. Engineered Canal Analysis

7.1 Chicago Sanitary and Ship Canal

The Chicago Sanitary and Ship Canal is a very appropriate site where the Great Bubble Barrier (GBB) should be implemented, especially as compared to the natural river environments studied in various states in the United States of America. In contrast to natural rivers, which are usually characterised by changes in the width, depth, and velocity of the flow, this engineered canal presents predictable and uniform hydraulic conditions directly related to the needs of the operation of the GBB system.

Structurally, the canal has a relatively constant breadth of about 50 metres, which falls within the upper limit to have sufficient success in forming a bubble curtain. Unlike this, large natural rivers like the Mississippi River or Susquehanna River can be much more than 300-500 metres wide, so that technically it would be impossible to have a continuous and effective bubble barrier across the entire width of the channel.

Another parameter that is important in GBB performance is depth. The Chicago Sanitary and Ship Canal has an average depth varying between 7 and 8 metres, which is far more than the recommended limit of 1 metre. This layer provides adequate vertical space for the formation of bubbles, and an uprise of air bubbles effectively transfers plastic debris to the surface. However, the natural rivers like the Lehigh River are seasonally low in water depth, especially during dry seasons, and this may drop below the operation limits and interrupt the operation of the system.

The canal and natural river systems also differ in flow velocity. Generally, the Chicago canal has a consistent velocity of flow that is in the range of between 0.4 and 0.5 metres per second, and this lies within the range of the optimal velocity that GBB can run under. The flow conditions have been considered stable so as to provide uniform tracks of the diagonal course of the bubbles, which steer the plastic debris to the collection point. However, natural rivers have high fluctuations of velocity as a result of seasonal floods, stormwater influx and melted snow. An example is when river currents, like those in the Hudson and Trinity, may pass at

speeds much faster than 1 metre/sec at high flow levels, leading to turbulence which interferes with bubble formation and capture effectiveness.

The other major benefit of the Chicago canal is that it is not subject to the changing water level. Designed canals exist to control the amount of water within an average range of less than 1 metre. This is essential to provide stable operation of the system and reduce structural stress to the installation. Conversely, natural rivers often undergo changes in the water level of more than 2 to 3 metres as a result of rainfall variability and seasonal hydrological cycles, which not only affect the structural robustness and working stability of GBB systems.

Besides hydraulic suitability, the Chicago Sanitary and Ship Canal also enjoys a steady and concentrated inflow of urban waste. The canal is a significant drainage and wastewater transportation system in the Chicago metropolitan region that has a population of over 2.7 million. This maintainable waste supply means that the GBB system is run in a high-impact environment where plastic interception can create quantifiable environmental outcomes.



Figure 21 Aerial view of the Chicago Sanitary and Ship Canal illustrating its engineered structure, uniform width, and controlled hydraulic conditions that support efficient bubble barrier deployment.

(Source: Wikipedia Contributors (2026) Chicago Sanitary and Ship Canal)

The canal has engineered infrastructure and is accessible, which further boosts operational feasibility. With the existence of well-developed maintenance systems and regulated routes of navigation, as well as predictable conditions of the flow, the complexity of the installation and cost of the operation in the long term are reduced. In comparison, natural rivers offer no logical simplicity in the form of channel morphology that does not conform to the elliptical channel, transport of sediments and ecological limitations that contribute to both maintenance and life of the system.

In general, the Chicago Sanitary and Ship Canal meets all important GBB requirements of a canal, such as width, depth, flow velocity, and the stability of water level on the surface. The discussion shows that, in contrast to natural river systems in various states, engineered canals deliver the hydraulic uniformity and operational dependability of an effective introduction of the GBB.

8. Comparative Feasibility Analysis

8.1 Criteria-Based Comparison Table

The comparison on the basis of the criteria is a clear indication that even the natural river systems can never hold the hydraulic demands said to allow efficient GBB to be implemented. Across the globe, rivers deposit around 1.15-2.41 million tonnes of plastic debris into oceans every year, which underlines the importance of rivers in pollution routes. Moreover, close to 1000 rivers are the source of almost 80 per cent of overall plastic emissions, which supports the necessity of efficient interception systems. Nonetheless, rivers that have been analysed in various states of the U.S are shown to be too wide, changes in the velocity of flows and variations in the seasons.

Criteria	Susquehanna River	Schuylkill River	Lehigh River	Los Angeles River	Hudson River	Chicago Canal
Width (≤ 50 m)	300–400 m	>100 m	Variable	>60 m	>150 m	~ 50 m
Depth (≥ 1 m)	Inconsistent	Canal only	Seasonal drop	Variable	Adequate	7–8 m
Velocity	High floods	Moderate	Variable	High	Strong	0.4–0.5

(<0.5 m/s)				flow	current	m/s
Stability	Flood & ice	Moderate	Unstable	Highly variable	Tidal influence	Controlled
Waste Interception	Partial	Low	Low	Low	Low	High
Overall Feasibility	Low	Low	Low	Low	Low	Very High

Criteria-Based Comparison Table

8.2 Critical Evaluation

The comparative study of river systems of nature and transport canals systems points to the basic difference of hydraulic stability, the possibility of working, and the efficacy of pollution interception. Rivers worldwide cause movement of heavy levels of plastic waste, with 0.8-2.7 million tonnes of plastic reported to flow into the oceans every year through the rivers. Besides, the latest research indicates that about 1.95 million metric tonnes of plastic pass through rivers annually, which is why river-based interception systems are very important.

Although this is a key factor in its operation, natural rivers pose significant technical challenges to the implementation of the Great Bubble Barrier (GBB). Channel width is one of the main issues since most of the large rivers are more than 100500 metres in width, which is way out of the operation limit of 50 metres needed to have a functional bubble curtain. Moreover, the velocity of the flow in natural rivers is quite inconsistent, often going beyond 0.5 m/s at the maximum discharge rates, and it is essential as it destabilises bubbles and minimized the effectiveness of plastic trapping. Instability is also caused by seasonal changes like flooding and drought, which lead to unstable performance of the system.

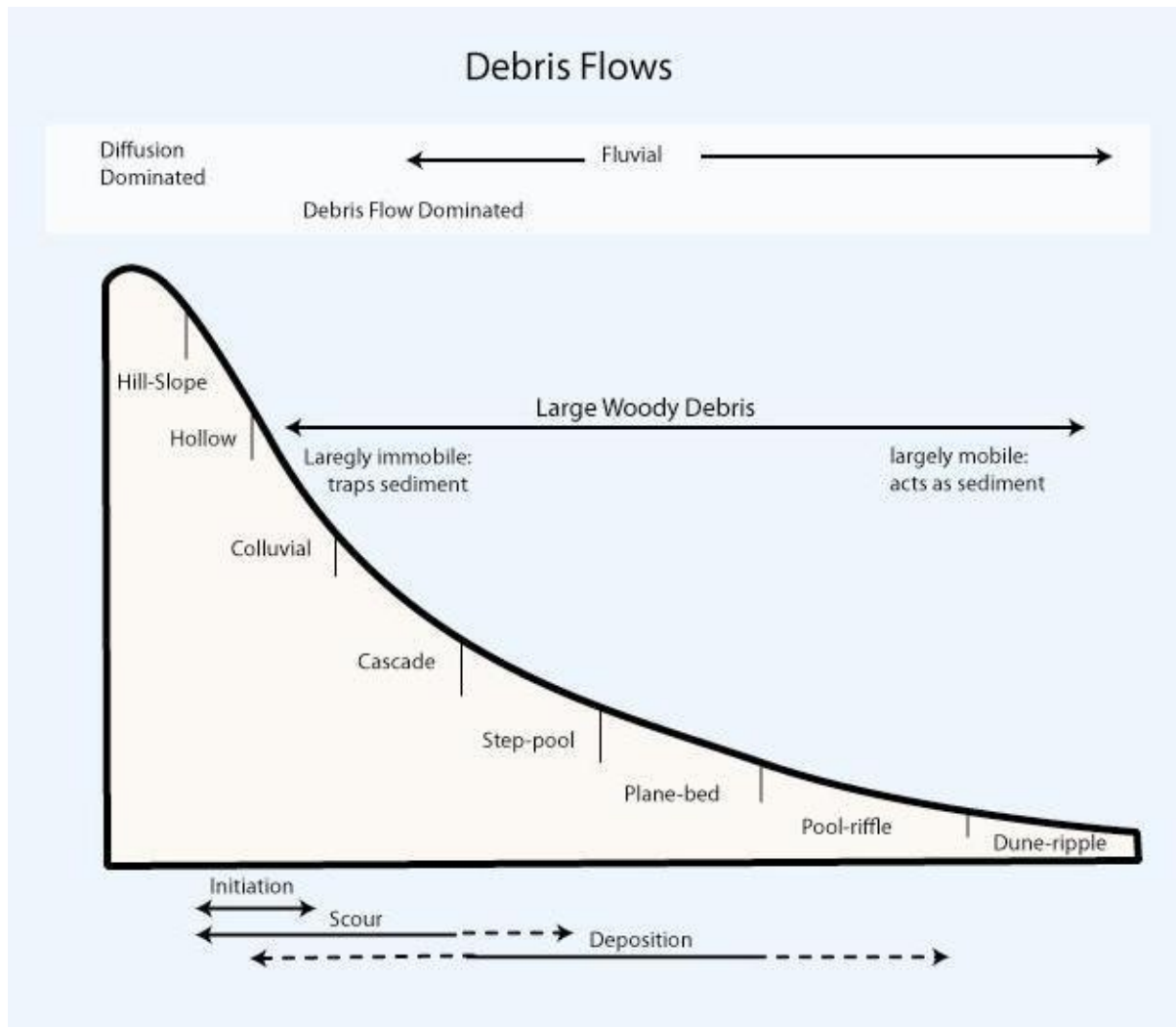


Figure 22 Comparative illustration of natural river systems and engineered canal systems, highlighting differences in flow stability, channel geometry, and suitability for controlled plastic interception technologies such as the Great Bubble Barrier.

(Source: McCabe, D. (2011))

However, engineered canal systems offer controlled hydraulic conditions that match GBB requirements regarding operation closely. Normal canals have a controlled width, a depth of more than 58 metres and a constant flow velocity that is at an optimal range of 0.40-0.50m/s. The conditions necessary to keep the formation of bubbles consistent and continuous are so that floating plastic waste can be effectively redirected and collected. Moreover, canal systems prediction is an advocacy of long-term dependability of operations and less intricate maintenance.

The other urgent difference is in the efficiency of garbage interception. Although natural rivers have the ability to transport vast amounts of plastic, their unpredictable flow patterns spread waste into broad channels, decreasing the performance of concentrated interception systems.

Canals, on the contrary, concentrate the waste in narrow corridors, enhancing the efficiency of capture and facilitating specific intervention (Politano *et al.*, 2024, August). Since more than 1923 million tonnes of plastic waste are released into water bodies each year, it is vital to focus on systems with a higher level of interception to result in a quantifiable environmental change.

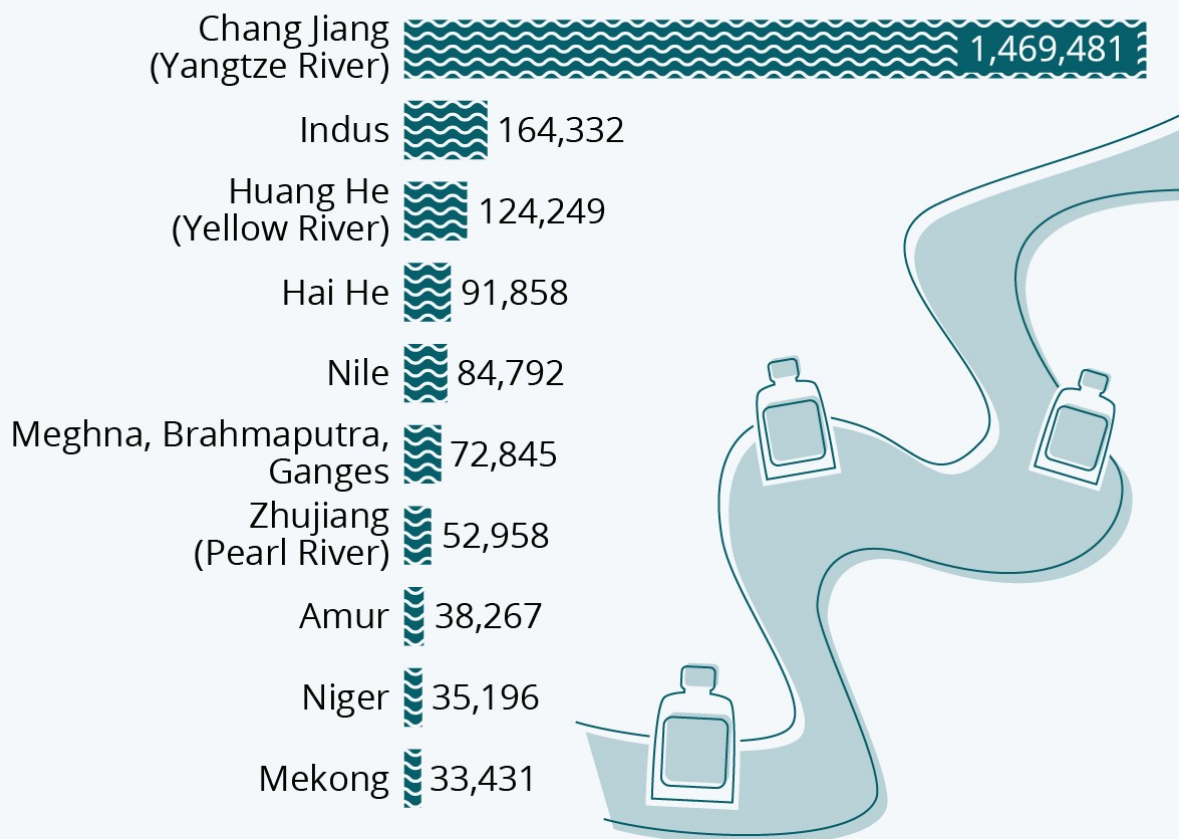
9. Discussion

9.1 Key Findings

According to the results, plastic pollution is a problem of great worldwide concern, and it is estimated that the amount of plastic that enters water bodies amounts to about 1923 million tonnes of plastic each year, and the rivers lose 0.827 million tonnes of plastic to the oceans. Moreover, approximately 1000 rivers serve approximately 80 per cent of plastic emissions, which implies the significance of a focused intervention. Analyses of natural rivers used in this paper were never able to satisfy important hydraulic needs, such as width, flow stability, and velocity constraints.

The Rivers Filling The Oceans With Plastic

Global rivers ranked by tonnes of plastic load carried into oceans each year*



* Data published in 2017.

Source: "Export of Plastic Debris by Rivers into the Sea" via United Nations



statista

Figure 23 Global distribution of river-based plastic pollution and its contribution to marine waste, highlighting the importance of interception technologies in reducing ocean-bound plastics.

(Source: McCarthy, N. (2021))

9.2 Environmental Implications

Plastic pollution poses a major environmental threat, especially in riverine and marine environments. Every year, about 1923 million tonnes of plastic waste finds its way into water,

negatively affecting the biodiversity and ecosystem stability in the world (Gambela, 2017). Rivers serve as a significant route of transport, and the world has approximately 1000 rivers, which lead to close to 80 per cent of plastic being released into oceans (Singh., 2025). Such degradation interferes with aquatic life, decreases water quality, and endangers more than 800 marine species with ingestion and entanglement. In turn, the successful interception mechanisms like the GBB are necessary to reduce the negative impact on ecology and promote the environmental management approaches at the international level.



Figure 24 Environmental impacts of plastic pollution on aquatic ecosystems, illustrating harm to marine species and habitat degradation caused by plastic accumulation.

(Source: Unfortunately, plastic bags. (2025))

9.3 Practical Challenges

There are a few practice challenges behind the implementation of the Great Bubble Barrier (GBB), namely the cost, scalability, and variability of the environment. Plastic pollution is an expanding problem across the world, with about 1923 million tonnes of plastic materials entering the water bodies each year, and this has caused a huge burden on mitigation systems. Implementation of GBB systems would demand a high initial cost, such as installation of the infrastructure, provision of compressor energy, and its maintenance over time (Press and Siqueira., 2022). At large-scale rivers, the high flow speeds and channel widths of over 50 m lead to reduced operational efficiency and high cost requirements of several installations. Also,

seasonal changes (flooding, sediment transport, and ice formation, etc.) may destabilise bubbles and lower their efficiency. Another challenge is scalability, as almost 80 per cent of worldwide plastic emissions are produced by more than 1000 rivers, which on a global scale would be resource-intensive to implement.



Figure 25 Environmental and hydraulic challenges affecting river-based plastic interception systems, including high flow velocity, sediment transport, and seasonal variability.

(Source: Plastic Pollution Crisis (2019))

10. Conclusion and Recommendations

10.1 Summary of Findings

The results mean that rivers are a major source of plastic pollution in the world, and the estimates suggest that 0.8-2.7 million tonnes of plastic get into oceans every year via rivers. Nonetheless, the feasibility study shows that the majority of natural rivers do not conform to the operational requirements of GBB, since they are so wide, their flow velocities are not constant, and their seasonal flows vary greatly. Engineered canals, on the other hand, offer predictable hydraulic situations, allowing the company to efficiently intercept and allowing the system to be reliably used in a predictable way, thus acting as a better solution to implement.

10.2 Final Recommendation

According to the assessment of the same, the deployment of the GBB in the United States should focus more on the engineered systems of canals, as compared to the natural rivers.

Rivers add 1.15 to 2.41 million tonnes of plastic material in the oceans each year, but the majority of natural systems are unable to achieve the area of hydraulic stability. Canals are designed to give a steady flow state in which performance remains constant, and all the wastes are intercepted effectively. Thus, the first application should be applied to urban canals to achieve the greatest environmental impact, operational stability and cost efficiency in the long run.

10.3 Future Research

The future research should aim at improving the GBB flexibility to non-uniform river environments, especially in high-flow and wide channel settings. Since tons of plastic have already been introduced to water bodies at a rate of 1923 million tonnes every year, there is a need to have efficient interception technologies with scaled-up capability. It should consider advanced modelling, real-time monitoring, and integration with smart waste management systems. As well, pilot studies with a large sample in different geographical locations are needed in order to evaluate the operational efficiency and economic feasibility over time.

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